Using the Tensile Tester

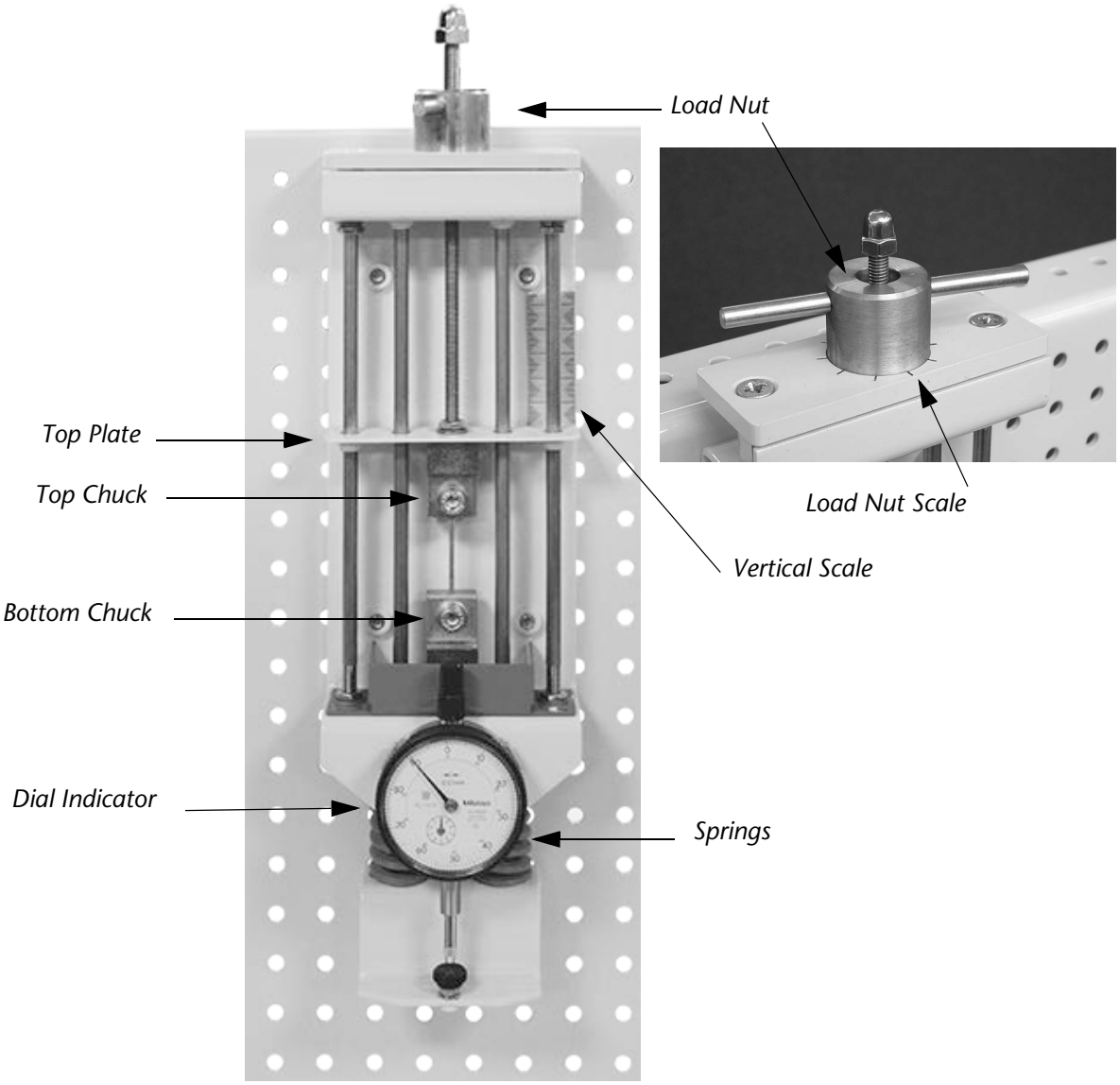


Figure 3 Front View of the Tensile Tester

The Tensile Tester is simple to use. Two chucks hold a specimen (see "Specimens" on page 10). The Load Nut at the top of the tester pulls the top chuck upwards, pulling the specimen up by 1 mm per complete turn. The Load Nut has a small notch and a scale divided into ten increments that helps you to see how far you have turned the control on each turn (each increment shows an upward pull of 0.1 mm).

The bottom chuck connects to the two large springs. A Dial Indicator measures the deflection of the springs as you apply force. The springs obey Hooke's Law, so the measurement from the Dial Indicator helps to show force.

The Top Plate works with a vertical scale to help you to remember how many turns you have done with the Force Control (one turn = 1 mm).

Fitting or Removing a Specimen

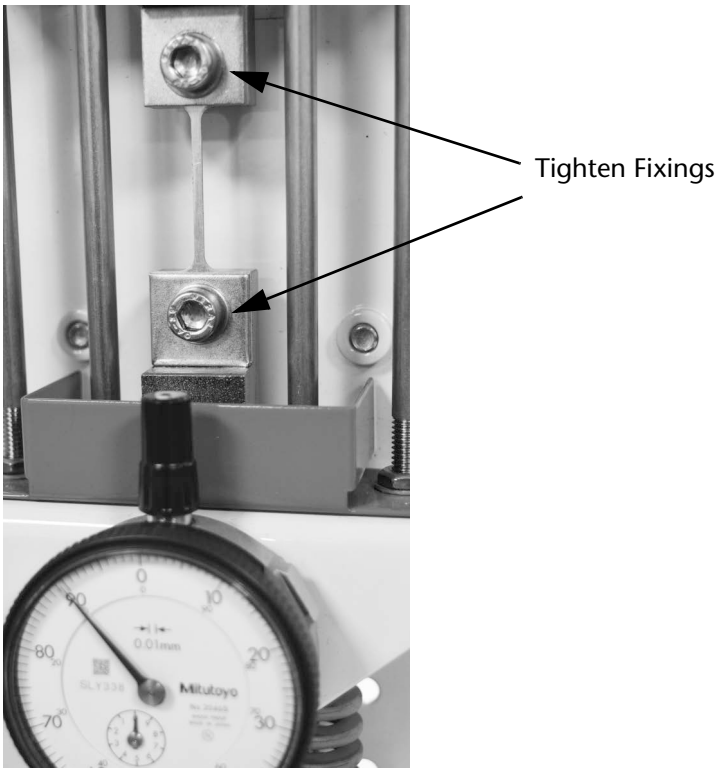


Figure 4 Use the Hexagon Tool (supplied) to Tighten Fixings

To fit a new specimen or remove a broken specimen, use the Hexagon Tool to undo the specimen fixings and refit them on the new specimen.

When fitting a new specimen, you may need to turn the Load Nut a few turns for the specimen to fit correctly.

Setting the Dial Indicator to Zero

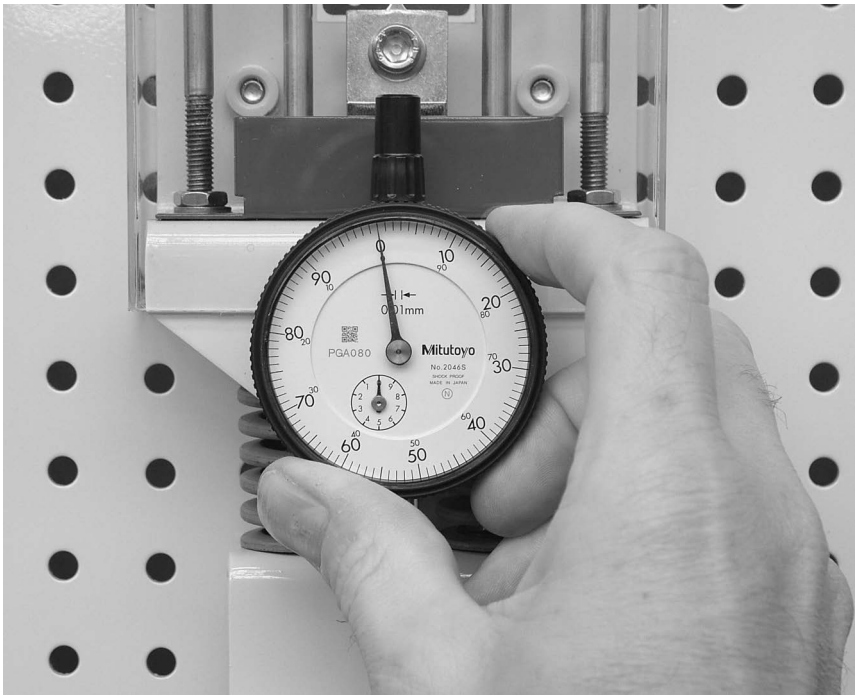


Figure 5 Turn the Outer Bezel

When you fit a new specimen, carefully turn the force control clockwise until the Dial Indicator just starts to show a movement and the notch of the Force Control points to one of the ten marks of its circular scale. Now turn the outer bezel of the dial indicator so that its pointer aligns with the 0 (zero) value.

Measuring Extension and Force

Extension (mm) = Number of Turns of Force Control (mm) - Dial Indicator Reading (mm)

For example - if you have turned the control three complete turns and the Dial Indicator reading is 2.83 mm, then the extension = 3.00 - 2.83 = 0.17 mm.

The Dial Indicator also gives an indication of the force applied to the specimen. The springs have a combined spring rate of 100 N/mm. The Dial Indicator gives a direct reading of the change in spring length, so each mm it changes by is equal to a change of force of one hundred Newton.

Force (N) = Dial Indicator reading (mm) x 100

For example - assuming you have correctly set the Dial Indicator to zero and it reads 2.83 mm after you apply the force, then the force is 283 N.

CAUTION 

Never try to test specimens other than those supplied by TecQuipment. You could break the tester.

Do not apply loads of greater than 1000 N.

Specimens

To test a particular material, it must be prepared in the form of a specimen. In real applications, specimens may be cut from a sample of a real part. For the tests with this kit, the specimens are specially made from different materials:

- A basic aluminium alloy
- Aluminium HE30 - a high specification aluminium alloy
- Mild steel - a basic steel alloy
- PVC (polyvinyl chloride) - a common thermoplastic

Aluminium HE30 and aluminium are very similar, so to help you, TecQuipment have put a small notch in the aluminium specimens. The steel specimens are heavier than the others and you can use a magnet (not supplied) to confirm they are made of steel. The PVC specimens are lighter in weight than the others and have a non metallic colour (usually black).

NOTE  *The PVC specimens may have a thin plastic film stuck to them, to help protect their surface from scratches. Remove this film before testing.*

TecQuipment make the specimens to fit in the machine. Each specimen has a large flat area at each end connected by a thin section. The machine holds the flat areas and stretches the thin section. This thin section is the 'gauge length'.

For your calculations, you will need to accurately measure the dimensions of your test specimens using the Dial Caliper supplied. Figure 6 shows the nominal dimensions for reference. To help easily compare results, all specimens have the same nominal dimensions.

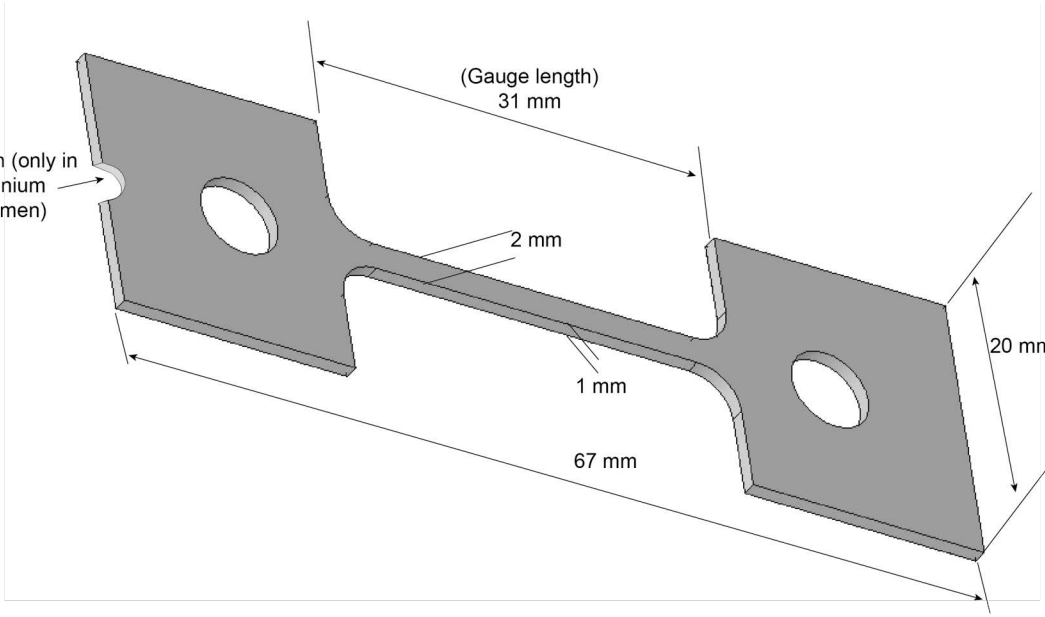


Figure 6 Nominal Specimen Dimensions

Theory

Notation

Symbol	Meaning	Units
^{o}	Area	m^2
F	Force	N
O	Stress	N/m^2 or Pa
J	Strain	-

Table 3 Notation

Stress (O)

Shown by Equation 1, this is the force applied to a material over a known area.

$$O = \frac{\text{Force}}{\text{Area acted on by the force}} \qquad (1)$$

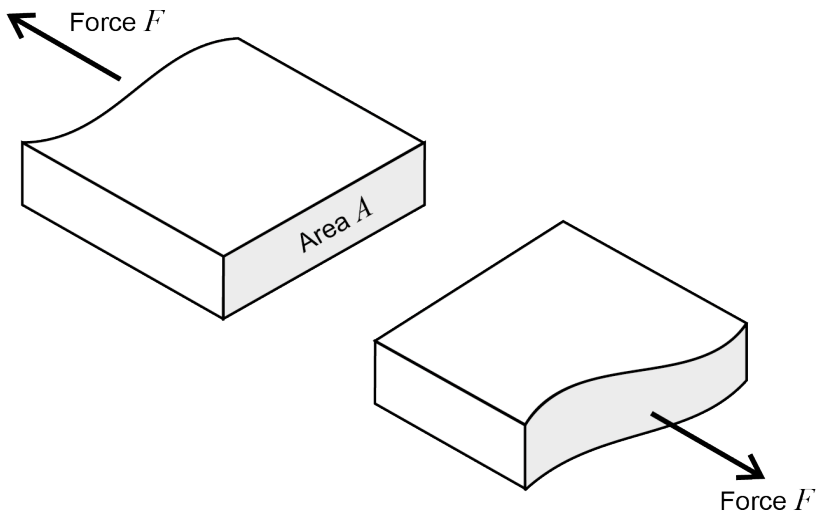


Figure 7 Force and Area

Compressive stress is where the material is compressed. It usually has a negative value.
Tensile stress is where the material is stretched. It usually has a positive value.

For the specimens in the kit, the area is the cross-sectional area of the ***thinnest part*** of the specimen along the gauge length, so:

$$O = \frac{F}{A} \qquad (2)$$

NOTE  *In reality, as a specimen stretches, its cross-sectional area decreases, which has a reciprocal affect on the stress. It is normal in tensile tests to assume a constant cross-sectional area.*

Units of stress are usually N/m^2 or Pa. Alternative divisions of these units are N/mm^2 and megapascals (MPa).

$$1 \text{ MPa} = 1 \text{ N/mm}^2$$

Strain (J)

Shown by Equation 3, this is the change in length (distortion caused by stress) of a material over its original length.

$$J = \frac{\text{Change in Length}}{\text{Original Length}} \qquad (3)$$

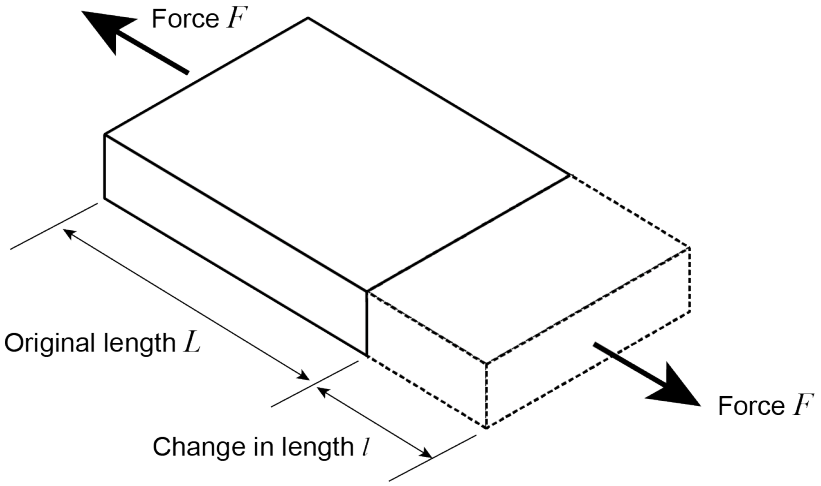


Figure 8 Change in Length

$$J = \frac{l}{L} \qquad (4)$$

Because strain is the ratio of two distances, it is dimensionless (it has no units). However, since measured strain is usually very small it is often shown in the form of 'microstrain' (I J) = strain x 10^{-6}.

Compressive strain is where the material has compressed. It usually has a negative value.
Tensile strain is where the material has stretched. It usually has a positive value.

Elongation and Percentage Elongation

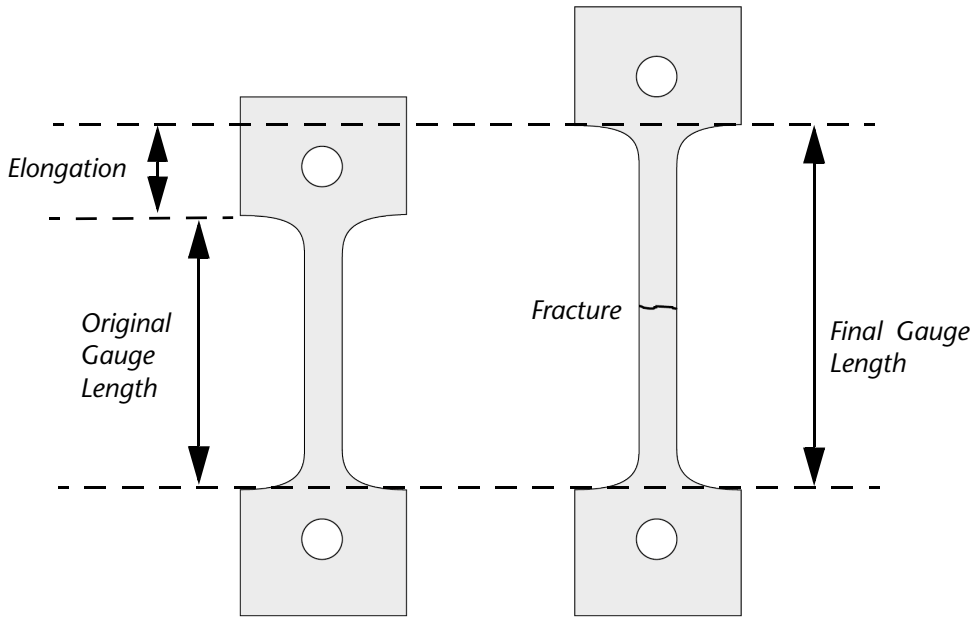


Figure 13 Elongation

Elongation is a simple value, often stated as an indication of ductility. You find it by subtracting the ***final*** length of the gauge length of a test specimen (***after it has broken***) by its original gauge length. You will need to carefully push the two pieces back together at the fracture point to measure the final length. The vertical scale on back of the tester should also give a reasonable indication of the elongation after fracture if you turn the force control back until the two pieces meet at the fracture.

The elongation is the difference between the final length and the original length.

$$\text{Elongation} = \text{Final Length} - \text{Original Length}$$

To calculate the value as a percentage:

$$\%\text{Elongation} = \frac{\text{Final Length} - \text{Original Length}}{\text{Original Length}} \cdot 100$$

Worked example:

Length of test section before test = 31 mm
Length after test = 33 mm
Elongation = 33-31 = 2 mm.
%Elongation = [(33-31)/31] x 100 = 6.45%

Tensile Tests

Aims

To investigate the tensile properties of different materials.

Experiment Setup

As shown in Figure 1, fix the Tensile Tester to the Work Panel:

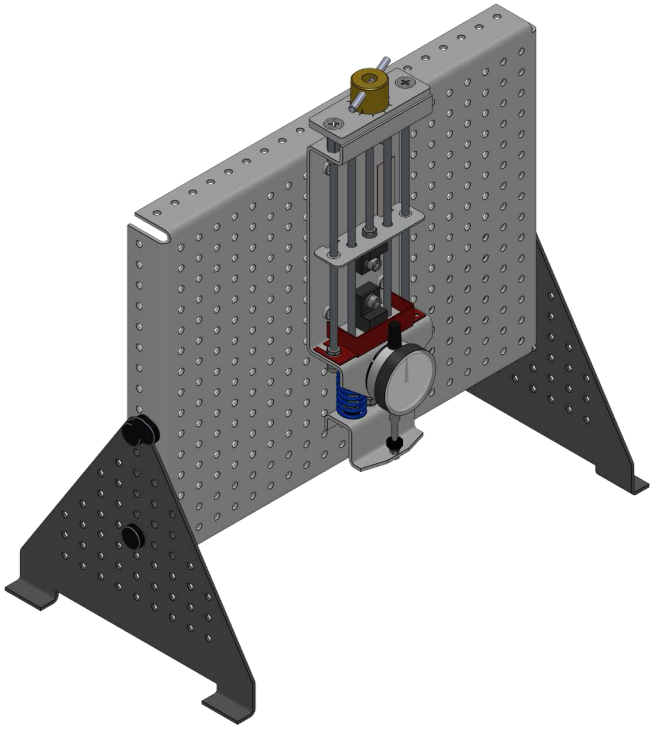


Figure 1 Experiment Setup

Experiment Procedure

1. Read the Guidance Notes (supplied).

2. Choose your specimen and use the Dial Caliper to measure the thickness and width at the gauge length and find its cross-sectional area. For reference, measure and record the original length.

3. Fit the specimen to the Tensile Tester and set the Dial Indicator to zero.

4. Note the reading on the Vertical Scale.

5. As shown in the blank Results Table, slowly turn the Load Nut clockwise in small 0.2 mm steps up to 5 mm, then larger 1 mm and 10 mm steps until the specimen breaks.

   Keep the applied rate consistent (for example: take five seconds to turn the Load Nut through each 0.2 mm and then take five seconds to record the reading).

   At each step, record the Dial Indicator value.

   NOTE 

   For the PVC specimens, after the yield point, the Dial Indicator reading slowly drops each time you turn the Load Nut, and may continue dropping for many minutes.

   For this reason, you must record the value as soon as you change the load to give consistent results.

   Metal specimens may not break until an extension of between 1 and 5 mm

   PVC specimens may not break until an extension of between 20 and 50 mm

6. To check the elongation, remove the specimen from the tester and push its broken ends together then measure its final length.

7. Convert the Dial Indicator readings into force values.

8. Subtract the Dial Indicator readings from the Load Nut movement readings to find the extension at each step.

9. Convert your force and extension values into stress and nominal strain values.

10. Plot your stress and strain results on the chart paper for your specimen (for best presentation, the steel, PVC and alloy specimens each have their own chart paper).

11. From your charts, note the yield point and tensile strength for each specimen.

12. For each specimen, find the gradient of the elastic region to compare the stiffness of the materials.

Questions and Conclusions

Two of the specimen materials should produce a similar gradient in the elastic region. Why is that?

Why do you think a constant force application rate and temperature are so important for PVC specimens?

Specimen material:
Original cross-sectional area (mm^2 or m^2):
Original Length (mm):

Load Nut movement (mm)	Dial Indicator (mm)	Force (N)	Extension (mm)	Stress O N/m^2	Nominal Strain J
0					
0.2					
0.4					
0.6					
0.8					
1.0					
1.2					
1.4					
1.6					
1.8					
2.0					
2.2					
2.4					
2.6					
2.8					
3.0					
3.2					
3.4					
3.6					
3.8					
4.0					
4.2					
4.4					
4.6					
4.8					
5.0					
6.0					
7.0					
8.0					
9.0					
10.0					
20.0					
30.0					
40.0					
50.0					
60.0					

Tensile Strength (N/m^2):
Yield Stress (N/m^2):
% Elongation:

Table 1 Blank Results Table

Specimen material:
Original cross-sectional area (mm^2 or m^2):
Original Length (mm):

Load Nut movement (mm)	Dial Indicator (mm)	Force (N)	Extension (mm)	Stress O N/m^2	Nominal Strain J
0					
0.2					
0.4					
0.6					
0.8					
1.0					
1.2					
1.4					
1.6					
1.8					
2.0					
2.2					
2.4					
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3.0					
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4.8					
5.0					
6.0					
7.0					
8.0					
9.0					
10.0					
20.0					
30.0					
40.0					
50.0					
60.0					

Tensile Strength (N/m^2):
Yield Stress (N/m^2):
% Elongation:

Table 2 Blank Results Table

Specimen material:
Original cross-sectional area (mm^2 or m^2):
Original Length (mm):

Load Nut movement (mm)	Dial Indicator (mm)	Force (N)	Extension (mm)	Stress O N/m^2	Nominal Strain J
0					
0.2					
0.4					
0.6					
0.8					
1.0					
1.2					
1.4					
1.6					
1.8					
2.0					
2.2					
2.4					
2.6					
2.8					
3.0					
3.2					
3.4					
3.6					
3.8					
4.0					
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8.0					
9.0					
10.0					
20.0					
30.0					
40.0					
50.0					
60.0					

Tensile Strength (N/m^2):
Yield Stress (N/m^2):
% Elongation:

Table 3 Blank Results Table

Specimen material:
Original cross-sectional area (mm^2 or m^2):
Original Length (mm):

Load Nut movement (mm)	Dial Indicator (mm)	Force (N)	Extension (mm)	Stress O N/m^2	Nominal Strain J
0					
0.2					
0.4					
0.6					
0.8					
1.0					
1.2					
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6.0					
7.0					
8.0					
9.0					
10.0					
20.0					
30.0					
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60.0					

Tensile Strength (N/m^2):
Yield Stress (N/m^2):
% Elongation:

Table 4 Blank Results Table

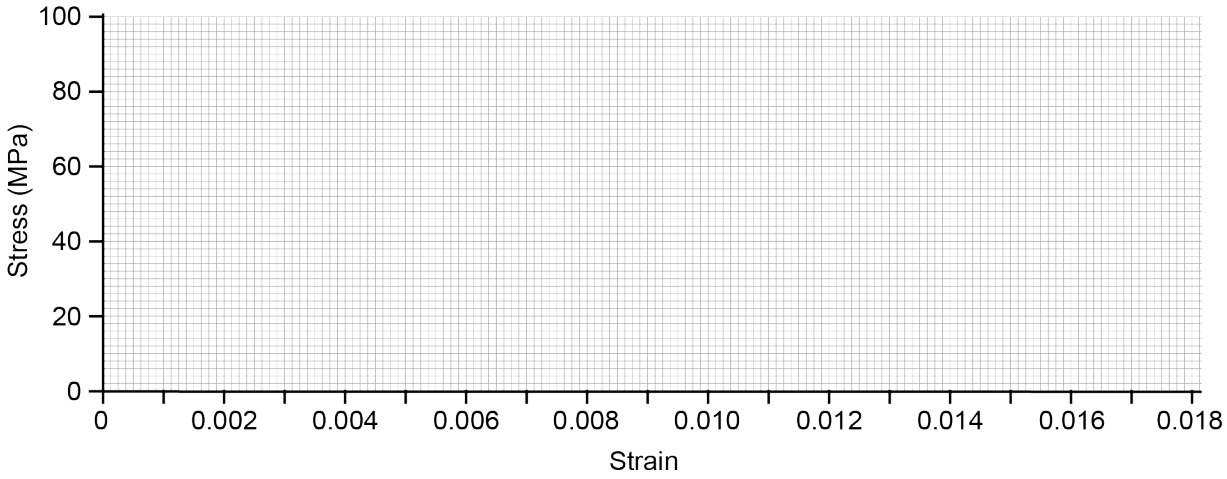


Figure 2 Blank Chart Paper - Aluminium Specimens

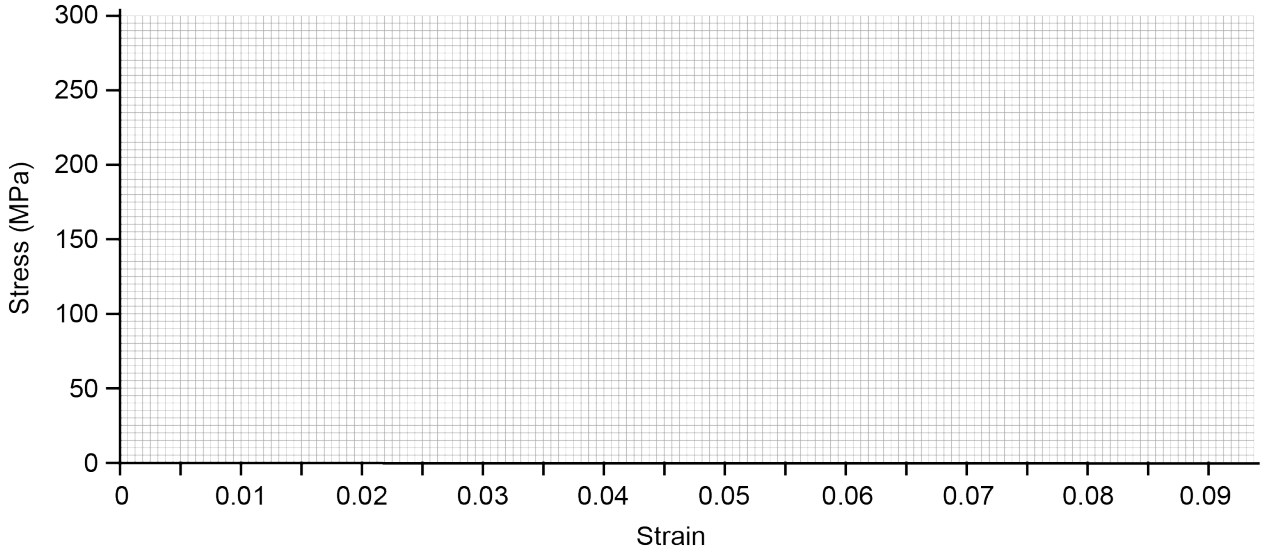


Figure 3 Blank Chart Paper - Aluminium HE30 Specimens

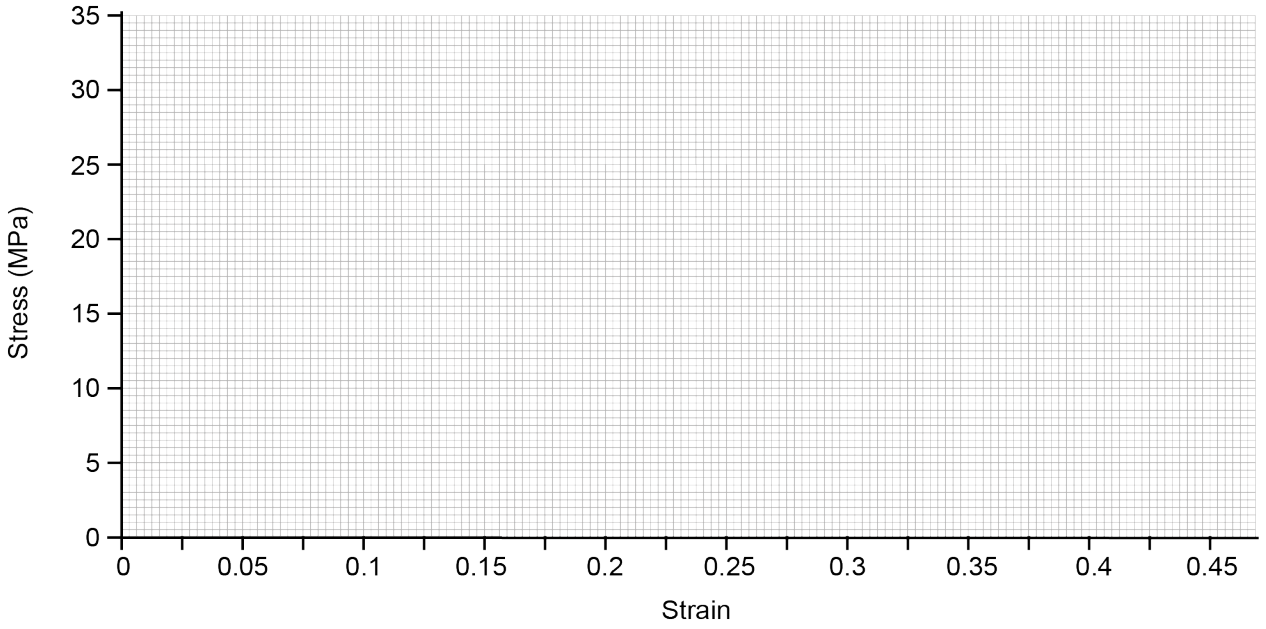


Figure 4 Blank Chart Paper - PVC Specimens

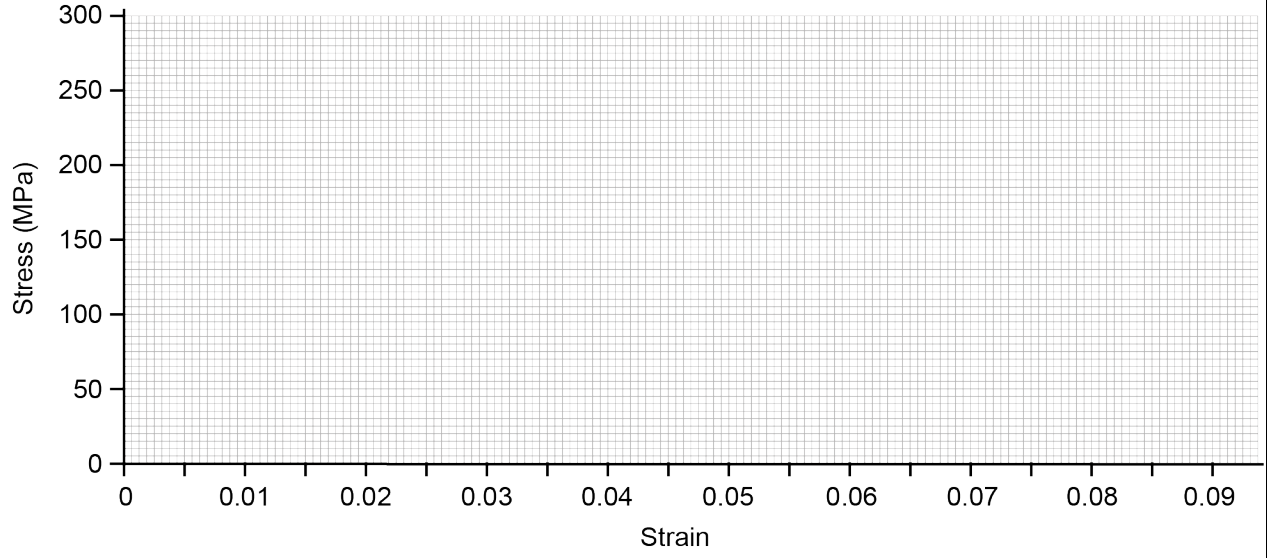


Figure 5 Blank Chart Paper - Steel Specimens